

Mustafa Ahmed Elnaiem

Senior Software Engineer · Systems Architect · EdTech Specialist

📍 Madinah, Saudi Arabia

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PROFESSIONAL SUMMARY

Senior Software Engineer and Systems Architect with an MSc in Computer Science and over 8 years of experience across software development, LMS administration, Linux server management, and database engineering. Specialized in ASP.NET Core Clean Architecture, Moodle plugin development, Oracle and MariaDB database administration, and production Linux deployment. Currently serving at Alrayan National Colleges, Madinah as part of the IT team — responsible for full-stack software development, security engineering, LMS administration, and end-to-end ownership of all systems I developed including deployment, database management, and infrastructure configuration. Co-founder and lead developer at Yulpana EdTech, building integrated school management systems deployed in production in West Africa. Author of a commercially available enterprise Alumni Management Portal deployed in production with 18 functional modules.

TECHNICAL SKILLS

Backend	ASP.NET Core (.NET 8), C#, Clean Architecture, CQRS, MediatR, PHP, Python
Frontend	Razor Pages, HTML5, CSS3, JavaScript, Mustache Templates, AMD Modules
Databases	Oracle (SIS integration), MariaDB, PostgreSQL, Microsoft SQL Server, EF Core
LMS	Moodle administration, Moodle plugin development (PHP), Koha library system (SIP2, LDAP)
Infrastructure	Ubuntu/Linux server admin, Nginx, Apache, UFW, fail2ban, systemd, shell scripting
Integration	LDAP/Active Directory, Oracle SIS, Telegram Bot API, Moodle Web Services, SignalR
DevOps	Linux deployment, Serilog, Hangfire, server hardening, backup automation
Architecture	Clean Architecture, CQRS, Vertical Slice, Microservices, REST API design
Languages	Arabic (Native), English (Professional Working Proficiency)

PROFESSIONAL EXPERIENCE

Software Engineer & IT Systems Administrator Alrayan National Colleges (AMC)

Nov 2021 — Present
Madinah, Saudi Arabia

- Designed and developed a production-grade Alumni Management Portal from scratch — 18 modules including LDAP authentication, Oracle SIS integration, configurable multi-step workflow engine, real-time SignalR notifications, PDF certificate generation, mock exam management with Moodle grade sync, jobs and recruitment portal, consultation booking, surveys, ratings, and full bilingual Arabic/English RTL support.
- Built a dedicated read-only SIS REST microservice bridging Oracle database to the Alumni Portal — enabling SIS portability across different student information systems with minimal configuration changes.
- Developed and maintained Moodle LMS for 3,000 students — administration, performance tuning, theme customization, and plugin development.
- Customized and deployed Koha library system including SIP2 protocol integration, LDAP authentication, UI overhaul, slideshow plugin, and multi-location visitor management system.
- Administered Oracle, MariaDB, and PostgreSQL databases — query optimization, backup automation, and performance tuning under production load
- Managed Ubuntu Linux production servers — Apache reverse proxy, UFW firewall hardening, fail2ban configuration, systemd service management, and automated shell script backups.
- Led incident response for a coordinated brute-force attack — implemented fail2ban with custom filters and automated IP ban scripts. Conducted full forensic investigation and remediation of a server compromise including webshell removal, reverse shell payload elimination, rogue SSH key cleanup, and malicious database entry removal
- Developed AcademicWorkflow system in ASP.NET Core Clean Architecture with LDAP authentication, role-based access control, and Linux subdirectory deployment

Co-Founder & Lead Developer

Jan 2023 — Present
Remote — NBS Mauritania

Yulpana EdTech (40% equity stake)

- Co-founded an EdTech startup building an integrated school management ecosystem — first live deployment at New Birth International School (NBS), Mauritania
- Developed a complete accounting module from scratch including chart of accounts, journals, general ledger, balance sheet, income statement, payroll processing, and cost centers
- Built a full transport management module with GPS integration, route management (home-to-school and school-to-home), vehicle and driver assignment, student boarding management, and real-time notifications
- Developed local_dashboards Moodle plugin — a comprehensive system with role-based dashboards for teachers, students, parents, and management; weekly plan template engine with auto-generation; Telegram bot integration with intelligent topic routing across 11 report types; gamification system; and trilingual support (English, Arabic, French)
- Customized Gibbon EduCore school management system and integrated it with Moodle LMS and Telegram for unified school operations

Lecturer — Computer Science University of Khartoum, Faculty of Engineering

Feb 2020 — May 2021
Khartoum, Sudan

- Delivered undergraduate lectures in computer science and software engineering subjects
- Developed and delivered course materials for programming and systems courses

EdTech Manager MSC Freelancing

Jan 2021 — Jul 2021
Remote

- Managed EdTech projects and digital learning implementations for educational clients

Software Engineer Hash for Information Technology

Jul 2019 — Mar 2020
Khartoum, Sudan

- Developed software solutions for enterprise clients — full-stack web development and system integration

Moodle Developer Innovative Solutions

Jan 2018 — Jul 2019
Khartoum, Sudan

- Developed Moodle plugins and customizations for educational institutions
- Built and administered Moodle LMS deployments — theme development, plugin integration, and user management

IT Lab Instructor The Future University

Nov 2016 — Nov 2017
Khartoum, Sudan

- Supervised computer science laboratory sessions and supported undergraduate students in practical coursework

EDUCATION

MSc in Computer Science

2018

University of Khartoum — Khartoum, Sudan

One of Sudan's most prestigious research universities

BSc in Computer Science

2016

The Future University — Khartoum, Sudan

KEY PROJECTS

Alumni Management Portal — ASP.NET Core · Clean Architecture · Oracle SIS · SignalR · Hangfire · Linux

Enterprise alumni management platform with 18 modules deployed in production at Alrayan National Colleges, Madinah. Features configurable workflow engine, LDAP/local authentication, real-time notifications, PDF generation, Moodle LMS integration, bilingual Arabic/English RTL, and SIS-agnostic microservice architecture. Available for licensing to other institutions.

local_dashboards — Moodle Plugin — PHP · Moodle API · Telegram Bot API · Scheduled Tasks

Production Moodle plugin with role-based dashboards, weekly plan management engine, Telegram bot integration with intelligent topic routing, 11 configurable report types, gamification system, and trilingual support. Deployed at NBS, Mauritania.

School Accounting & Transport System — PHP · Gibbon EduCore · GPS Integration

Full accounting module (GL, journals, payroll, cost centers) and GPS-integrated transport management system built for NBS as part of the Yulpana school management ecosystem.

Koha Library System Customization — Perl · SIP2 · LDAP

Full UI overhaul, SIP2 protocol integration, LDAP authentication, recent books slideshow plugin, and multi-location visitor management system deployed at AMC.

REFERENCE CLIENT

Alrayan National Colleges (AMC) — Madinah, Kingdom of Saudi Arabia

Live production deployment of Alumni Management Portal, Moodle LMS, Koha library system, and AcademicWorkflow system. Available as reference upon request.